# Thesis erratum

**Thesis.** Minjun Kim, *Electrical measurements and prediction of dielectric barrier surface damage in nonthermal atmospheric pressure dielectric barrier discharge plasma based on machine learning*, M.S. thesis, Kwangwoon University, 2023.

**Date of this erratum.** 2026-11-01.

**Basis.**

- Each item refers to an entry of the verification report, given by its ID (A1, C5, …).
- "Re-analysis" values come from device m2 (40 normal and 120 degraded captures).

**Page numbers.** These are the numbers printed on the thesis pages. The thesis is written in Korean, so the original wording is quoted in Korean only in the Korean version.

**Types of item.**

| Type | Meaning |
|---|---|
| **Correction** | Error of fact, arithmetic or definition |
| **Addition** | Missing interpretation or limitation |
| **Notation** | Units, typos, formatting |

> The thesis PDF itself is unchanged. Read it together with this erratum and the verification report.

## Abstract (pp. v–vi)

| # | Location | Correction | Type · ref |
|---|---|---|---|
| 1, 3 | KO and EN abstract | Replace "discharge energy per second" with "**discharge energy per cycle**". | Correction · C1 |
| 2, 4 | KO and EN abstract, sentence on changes due to damage | Replace with: "Surface damage decreased the effective voltage, effective current, discharge-onset voltage and burning voltage, and increased the capacitances (C_total, C_gas). The discharge energy, number of discharges and electron density did not change significantly." | Correction · D1 |

## Chapter 1

| # | Page | Correction | Type · ref |
|---|---|---|---|
| 5 | 1 | Write "electron temperature of a few eV (≈ $10^4$–$10^5$ K), gas temperature ≈ 300 K" instead of "> 2000 °K / 300 °K". | Notation · A5 |
| 6 | 1 | Add: "electron density in atmospheric DBD micro-discharge channels is $10^{14}$–$10^{15}$ $cm^{-3}$ [61]". The quoted $10^{10}$–$10^{13}$ $cm^{-3}$ applies to low-pressure plasmas. | Addition · A5 |
| 7 | 3 | Eq. (7) is **Townsend's semi-empirical formula**, not the Saha ionisation formula. | Correction · A1 |
| 8 | 5 | Below the Paschen curve there is no breakdown. To the right of the minimum, collisions are too frequent for electrons to gain the ionisation energy; to the left, there are too few collisions. | Correction · A4 |
| 9 | 6 | For a 100 µm air gap, Eq. (8) gives a breakdown voltage of ≈ 0.7–0.9 kV, not "several kV". The applied voltage must reach ≈ 2 kV peak because of the capacitive divider formed by the dielectric. | Correction · A3 |

## Chapter 2

| # | Page | Correction | Type · ref |
|---|---|---|---|
| 10–11 | 11 | The HV probe is the **P6015A**; the current probe is the **P6021A**. | Correction · B1 |
| 12 | 11–12 | Add the measurement limits: sampling 10 ns (some captures 20 ns), 5000 points (≈ 2.3 periods), 8 bit (40 V/LSB; 1.6–8 mA/LSB), current-probe bandwidth 60 MHz. As a result, pulse counts and peak currents are lower bounds. | Addition · B2 |

## Chapter 3

| # | Page | Correction | Type · ref |
|---|---|---|---|
| 13 | 16 | V_min is ≈ 300–700 V (mean 504.5 ± 168.8 V), not "300–500 V". | Notation · C3 |
| 14, 16 | 16–17 | "Energy per second" → "**energy per cycle**": 0.13 ± 0.022 mJ (normal) and 0.13 ± 0.024 mJ (degraded). Add the mean power P = E·f ≈ **6.0 W**. | Correction · C1 |
| 15, 34 | 16, 32–33 | Write mean ± SD in a single unit: 1.50 ± 0.017 kV, 20.8 ± 4.0 mA, 34.2 ± 4.9 pF, 11.3 ± 1.2 pF, 17.3 ± 2.8 pF, 59.4 ± 9.0 nC, etc. | Notation · C10 |

| # | Page | Correction | Type · ref |
|---|---|---|---|
| 17 | 17 | Eq. (11) → **$E = \int_0^T v\, i\, dt$** [J/cycle] and P = E/T [W]. | Correction · C1 |
| 18–20 | 19–21 | Replace Eqs. (13)–(15) with **text A** below. 55.6 nC / 30.5 pF = 1.82 kV, not 1.5 kV. The original Eq. (15) evaluates to V_max (2.1/2.0 kV). The burning voltage is U_b = 860 ± 68 V (normal) and 811 ± 75 V (degraded). | Correction · C4, C5 |
| 21 | 22 | Describe the pulse detector exactly: threshold = median prominence × 1.5, and pulses with IQR-outlier widths removed. Note that this threshold is relative to each capture. | Addition · C11 |
| 22 | 23 | Fig. 14 shows $T_3$–$T_4$, not $T_1$–$T_2$. | Notation · C3 |
| 23, 25 | 24–25 | In Eqs. (18) and (20), use **U_b** instead of V_breakdown. n_e then becomes ≈ $3.9\times10^{12}$ (normal) and $4.0\times10^{12}$ $cm^{-3}$ (degraded), with no significant difference. Subtracting the displacement current from i_plasma is recommended. | Correction · C7 |
| 24 | 24 | The area $4.5\times10^{-4}$ $cm^2$ is the pad top area, used as an approximation of the current cross-section. n_e is therefore a cell-averaged relative index. | Addition · C7 |
| 26 | 26 | Use N = p/(k_BT) = $2.50\times10^{19}$ $cm^{-3}$ (1 atm, 20 °C). This is the gas number density, not "Avogadro's number". | Correction · C6 |
| 27 | 26 | E/N ≈ $34\times10^{-16}$ V $cm^2$ (≈ 340 Td) with U_b. Note that even the original inputs give 75.0, not 81.38. | Correction · C6 |
| 28 | 27 | Within the data range of Fig. 15, D/μ ≈ 6 V, so T_e ≈ **6 eV**. Re-analysis: 6.2 ± 0.3 eV (normal), 6.0 ± 0.4 eV (degraded). The original 13 eV was read from the extrapolated part of the curve. | Correction · C6 |
| 29 | 26 | Add: D/μ is the characteristic energy; it equals k_BT_e/e only for a Maxwellian distribution. | Addition · C6 |
| 30 | 28 | Eq. (23) → **$Q(t) = \int_0^t i\, dt' + Q_0$**, after removing the current offset. | Correction · C2 |
| 31–33 | 32 | Table 2: the C_total fitting error is $7.63\times10^{-14}$ F. Write $p < 10^{-15}$ instead of $1.11\times10^{-16}$. Add $R^2$ ≈ 0.98 and note the autocorrelation of the fitted points. | Correction · C8 |
| 35 | 32 | Add: the discharge-phase slope is an effective ζ_diel ≤ C_d when the discharge covers only part of the dielectric [50]. | Addition · C9 |

**Text A** (replaces Sec. 3.1.3, Eqs. 13–15)

- The gas-gap voltage is u_g(t) = v(t) − Q(t)/C_dielectric (13′). During discharge it stays approximately at the burning voltage U_b [42, 55].
- At ignition, the wall charge Q_on left over from the previous half cycle is still on the dielectric. It produces V_wall = −Q_on/C_dielectric (14′).
- Hence U_b = V_min − Q_on/C_dielectric = (1 − C_total/C_dielectric) · (V_max + V_min)/2 (15′).

- With the mean values of Table 2, $U_b \approx 0.92$ kV, consistent with the Paschen value for a 100 µm air gap (0.72–0.87 kV). Re-analysis gives 860 ± 68 V (normal) and 811 ± 75 V (degraded).
- Why the original formula fails: using the transferred charge $Q_{max} = C_{dielectric}(V_{max} - V_{min})$ as the wall charge gives $V_{min} + Q_{max}/C_{dielectric} = V_{max}$.

## Chapters 4–5

| # | Page | Correction | Type · ref |
|---|---|---|---|
| 36 | 40 | "Degradaion" → "Degradation". | Notation |
| 37 | 43 | "Freqency" → "Frequency"; the unit of the electron-density SD is $cm^{-3}$. | Notation |
| 38 | 43 | Add: the features are not independent. $C_{gas}$ follows from $C_d$ and $C_{total}$; $T_e$ and $n_e$ follow from the breakdown voltage. | Addition · D9 |
| 39 | 44–45 | Remove the references to a DNN (no results are reported). | Notation · D9 |
| 40 | 45 | Add: the training/test set is a random split of time-ordered captures from one device, so test scores are optimistic. The cross-device validation set gives the primary performance. | Addition · D5 |
| 41 | 47 | "min_samples_splite" → "min_samples_split". | Notation |
| 42–44 | 49–50 | Precision = fraction of samples predicted *degraded* that are degraded. Recall = fraction of truly degraded samples that are predicted degraded. TP, TN, FP, FN refer to the classes degraded (positive) and normal (negative); **FN = degraded predicted as normal**. | Correction · D3 |
| 45 | 51 | Table 6: precision and recall are **macro averages**; F1 is the **binary F1** of the degraded class. | Correction · D2 |
| 46 | 51 | Add: on the validation set, LR and SVM reach AUC 0.99 but only 76.4 % accuracy. This is a threshold shift between devices, which per-device baseline normalisation removes. | Addition · D4 |
| 47 | 52 | Add: $V_{rms}$ alone almost separates the classes (AUC ≈ 1.0); the classifier mainly detects the drop of the applied voltage. | Addition · D6 |
| 48 | 53 | A random forest **averages** the votes of all 1000 trees; it does not select the best tree. Fig. 29 shows one tree. | Correction · D7 |
| 49 | 55 | Tables 7–8 show no drop of the "normal" probability before damage (0.886 / 0.909; 0.816 / 0.818). The "≥ 10 % drop" precursor criterion is therefore not supported and should be stated as future work. | Correction · D8 |
| 50 | 56 | "Fig. 32 (a)/(b)" → "Fig. 30 (a)/(b)". | Notation |

## Chapter 6 (p. 57)

| # | Correction | Type · ref |
|---|---|---|
| 51–52 | Same changes as items 1–2 (energy per cycle; the list of significant and non-significant changes). | Correction · D1 |

## References

| # | Correction |
|---|---|
| 53 | [25] duplicates [10] (Michel et al., J. Appl. Phys. 105, 073302 (2009)); merge them. |
| 54–55 | In [26] and [27], remove the affiliation letters from the author names: Gimeno, Hanson, Goldhawk, Simpson, Lavoie. |
| 56 | [28]: add the journal, *J. Appl. Phys.* 124, 183301 (2018). |
| 57 | [13]: add the title, journal, volume, pages and year. |
| 58 | [1]: check the edition (Plenum 2nd ed. 1984, or Springer 3rd ed. 2016). |